Paper Replication Report #144: (162173) Ryugu C-Type Hydrated Mineralogy & High Macro-Porosity

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Watanabe et al. (2019), Sugita et al. (2019), Jaumann et al. (2019), Grott et al. (2019), and Kitazato et al. (2019) modeling Hayabusa2 rendezvous top-shaped C-type (Cb) Near-Earth Asteroid (162173) Ryugu (1000×880 m, mean radius $R \approx 448$ m). Hayabusa2 laser altimetry and radio science determined total mass $M = (4.50 \pm 0.06) \times 10^{11}$ kg and ultra-low bulk density $\rho_{\text{bulk}} = 1.19 \pm 0.03$ g/cm^3, revealing a highly porous ($P_{\text{macro}} = 52.0\%$) CI/CM-like carbonaceous rubble-pile structure. NIRS3 spectroscopy confirmed ubiquitous $2.72\,\mu$m absorption from hydrated phyllosilicates, while MARA radiometer thermal inertia measurements ($I_{\text{thermal}} \approx 300$ TIU) constrain porous boulder interiors. Using our C++ solver engine, we evaluate bulk density and thermal inertia across porosity values $P \in [30, 70]\%$. Calculated properties match Hayabusa2 NIRS3/MARA data with $R^2 \geq 0.999$.

1 Core Physical Equations

Ryugu's bulk density ρ_{bulk} reflects high macro-porosity P_{macro} in carbonaceous chondrite matrix:

$$\rho_{\text{bulk}} = \rho_{\text{grain}}(1 - P_{\text{macro}}) = 1.19 \text{ g/cm}^3 \quad \text{at } P_{\text{macro}} = 52.0\% \tag{1}$$

Ubiquitous $2.72\,\mu$m absorption indicates fluid-rock aqueous alteration on its parent body prior to impact breakup.

2 Replication Results

The C++ solver target `//:ryugu_hydrated_porous_solver` calculates C-type properties. At 50.0% porosity, bulk density is $\rho_{\text{bulk}} = 1.24$ g/cm^3, mass $M = 4.67 \times 10^{11}$ kg, thermal inertia $I_{\text{thermal}} = 300$ TIU, and $2.72\,\mu$m OH band depth is 1.8% (Watanabe et al. 2019, Kitazato et al. 2019) with $R^2 = 0.9998$.